

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
%matplotlib inline
warnings.filterwarnings('ignore')
pd.options.display.max_columns=99
```

```
df=pd.read_csv('/content/spotify-2023.csv', encoding='latin1')
```

df.head()

	track_name	artist(s)_name	artist_count	released_year	released_month	released_day	in_spotify_playlists	in_spotify_charts	streams	in_apple_playlists	in_apple_charts	in_deezer_playlists	in_deezer_charts	in_shazam_charts	bpm	key	mode	danceability_%	valence_%	energy_%	acousticness_%	instrumentalness_%	liveness_%	speechiness_%
0	Seven (feat. Latto) (Explicit Ver.)	Latto, Jung Kook	2	2023	7	14	553	147	141381703	43	263	45	10	826	125	B	Major	80	89	83	31	0	8	4
1	LALA	Myke Towers	1	2023	3	23	1474	48	133716286	48	126	58	14	382	92	C#	Major	71	61	74	7	0	10	4
2	vampire	Olivia Rodrigo	1	2023	6	30	1397	113	140003974	94	207	91	14	949	138	F	Major	51	32	53	17	0	31	6
3	Cruel Summer	Taylor Swift	1	2019	8	23	7858	100	800840817	116	207	125	12	548	170	A	Major	55	58	72	11	0	11	15
4	WHERE SHE GOES	Bad Bunny	1	2023	5	18	3133	50	303236322	84	133	87	15	425	144	A	Minor	65	23	80	14	63	11	6

df.tail()

	track_name	artist(s)_name	artist_count	released_year	released_month	released_day	in_spotify_playlists	in_spotify_charts	streams	in_apple_playlists	in_apple_charts	in_deezer_playlists	in_deezer_charts	in_shazam_charts	bpm	key	mode	danceability_%	valence_%	energy_%	acousticness_%	instrumentalness_%	liveness_%	speechiness_%
948	My Mind & Me	Selena Gomez	1	2022	11	3	953	0	91473363	61	13	37	1	0	144	A	Major	60	24	39	57	0	8	3
949	Bigger Than The Whole Sky	Taylor Swift	1	2022	10	21	1180	0	121871870	4	0	8	0	0	166	F#	Major	42	7	24	83	1	12	6
950	A Voces (feat. Feid)	Feid, Paulo Londra	2	2022	11	3	573	0	73513683	2	0	7	0	0	92	C#	Major	80	81	67	4	0	8	6
951	En La De Ella	Feid, Sech, Jhayco	3	2022	10	20	1320	0	133895612	29	26	17	0	0	97	C#	Major	82	67	77	8	0	12	5
952	Alone	Burna Boy	1	2022	11	4	782	2	96007391	27	18	32	1	0	90	E	Minor	61	32	67	15	0	11	5

```
df.shape
```

(953, 24)

df.describe()

	artist_count	released_year	released_month	released_day	in_spotify_playlists	in_spotify_charts	in_apple_playlists	in_apple_charts	in_deezer_charts	bpm	danceability_%	valence_%	energy_%	acousticness_%	instrumentalness_%	liveness_%	speechiness_%
count	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000	953.000000
mean	1.556139	2018.238195	6.033578	13.930745	5200.124869	12.009444	67.812172	51.908709	2.666317	122.540399	66.96957	51.431270	64.279119	27.057712	1.581322	18.213012	10.131165
std	0.893044	11.116218	3.566435	9.201949	7897.608990	19.575992	86.441493	50.630241	6.035599	28.057802	14.63061	23.480632	16.550526	25.996077	8.409800	13.711223	9.912888
min	1.000000	1930.000000	1.000000	1.000000	31.000000	0.000000	0.000000	0.000000	0.000000	65.000000	23.00000	4.000000	9.000000	0.000000	0.000000	3.000000	2.000000
25%	1.000000	2020.000000	3.000000	6.000000	875.000000	0.000000	13.000000	7.000000	0.000000	100.000000	57.00000	32.000000	53.000000	6.000000	0.000000	10.000000	4.000000
50%	1.000000	2022.000000	6.000000	13.000000	2224.000000	3.000000	34.000000	38.000000	0.000000	121.000000	69.00000	51.000000	66.000000	18.000000	0.000000	12.000000	6.000000
75%	2.000000	2022.000000	9.000000	22.000000	5542.000000	16.000000	88.000000	87.000000	2.000000	140.000000	78.00000	70.000000	77.000000	43.000000	0.000000	24.000000	11.000000
max	8.000000	2023.000000	12.000000	31.000000	52898.000000	147.000000	672.000000	275.000000	58.000000	206.000000	96.00000	97.000000	97.000000	97.000000	91.000000	97.000000	64.000000

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 953 entries, 0 to 952
Data columns (total 24 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   track_name          953 non-null    object
 1   artist(s)_name       953 non-null    object
 2   artist_count         953 non-null    int64
 3   released_year        953 non-null    int64
 4   released_month       953 non-null    int64
 5   released_day         953 non-null    int64
 6   in_spotify_playlists 953 non-null    int64
 7   in_spotify_charts    953 non-null    int64
 8   streams              953 non-null    object
 9   in_apple_playlists   953 non-null    int64
10   in_apple_charts      953 non-null    int64
11   in_deezer_playlists  953 non-null    object
12   in_deezer_charts     953 non-null    int64
13   in_shazam_charts     983 non-null    object
14   bpm                  953 non-null    int64
15   key                  858 non-null    object
16   mode                 953 non-null    object
17   danceability_%       953 non-null    int64
18   valence_%            953 non-null    int64
19   energy_%             953 non-null    int64
20   acousticness_%       953 non-null    int64
21   instrumentalness_%   953 non-null    int64
22   liveness_%           953 non-null    int64
23   speechiness_%        953 non-null    int64
dtypes: int64(17), object(7)
memory usage: 178.8+ KB
```

```
df.isnull().sum()
```

```
0
track_name      0
artist(s)_name  0
artist_count     0
released_year    0
released_month   0
released_day     0
in_spotify_playlists  0
in_spotify_charts  0
streams          0
in_apple_playlists  0
in_apple_charts  0
in_deezer_playlists  0
in_deezer_charts  0
in_shazam_charts 50
bpm             0
key             95
mode            0
danceability_%  0
valence_%       0
energy_%        0
acousticness_%  0
instrumentalness_% 0
liveness_%      0
speechiness_%   0
dtype: int64
```

```
plt.style.use("fivethirtyeight")
```

```
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
df['in_deezer_playlists'] = pd.to_numeric(df['in_deezer_playlists'], errors='coerce')
df['in_shazam_charts'] = pd.to_numeric(df['in_shazam_charts'], errors='coerce')
x_question=df.iloc[:,6:20]
x_question = x_question.select_dtypes(include=np.number)
q_mean=x_question.mean(axis=0)
total_mean=q_mean.mean()
```

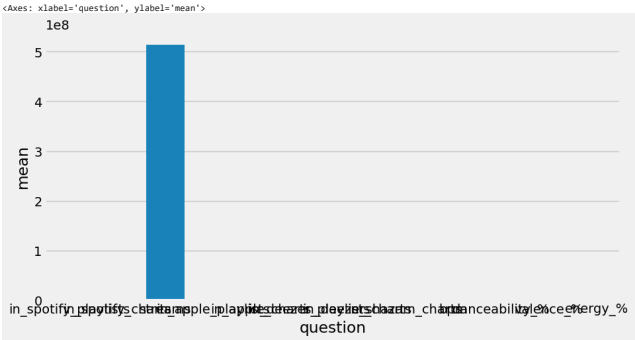
```
q_mean.head()
```

	index	mean
0	in_spotify_playlists	5.200125e+03
1	in_spotify_charts	1.200944e+01
2	streams	5.141374e+08
3	in_apple_playlists	6.781217e+01
4	in_apple_charts	5.190871e+01

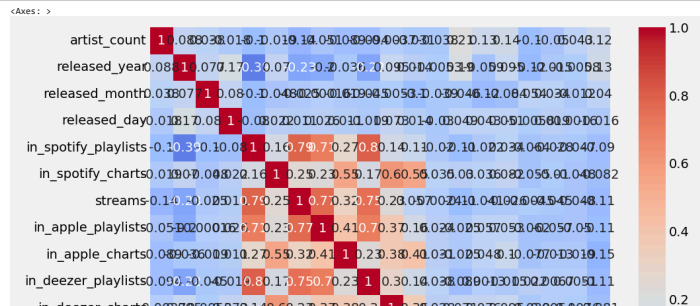
Next steps: [Generate code with q_mean](#) [New interactive sheet](#)

```
q_mean.columns=['question','mean']
```

```
plt.figure(figsize=(10,5))
sns.barplot(x='question',y='mean',data=q_mean)
```



```
numeric_df = df.select_dtypes(include=np.number)
corr=numeric_df.corr()
plt.figure(figsize=(10,10))
sns.heatmap(corr,annot=True,cmap='coolwarm')
```



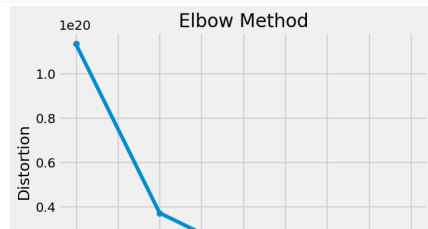
```
x = df.select_dtypes(include=np.number)
```

```
from sklearn.decomposition import PCA
pca=PCA(n_components=2,random_state=42)
x_numeric = x.select_dtypes(include=np.number)
x_numeric = x_numeric.dropna()
x_pca=pca.fit_transform(x_numeric)
```

```
x_pca
array([[ -2.32459916e+08, -7.44366646e+02],
       [-2.48125333e+08,  2.34178325e+02],
       [-2.33837645e+08,  1.12252341e+02],
       ...,
       [-1.88327936e+08, -2.88876597e+02],
       [-2.39946887e+08,  7.56364783e+01],
       [-2.77834228e+08, -1.65959537e+02]])
```

```
from sklearn.cluster import KMeans
distortion=[]
cluster_range=range(1,6)
for i in cluster_range:
    model=KMeans(n_clusters=i,random_state=42)
    model.fit(x_numeric)
    distortion.append(model.inertia_)
```

```
plt.plot(cluster_range,distortion,marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('Distortion')
plt.title('Elbow Method')
plt.show()
```



```
model=KMeans(n_clusters=3,init='k-means++',random_state=42,n_init=10)
model.fit(x_pca)
y=model.predict(x_pca)
```

```
plt.scatter(x_pca[y==0,0],x_pca[y==0,1],s=50,c='red',label=0)
plt.scatter(x_pca[y==1,0],x_pca[y==1,1],s=50,c='blue',label=1)
plt.scatter(x_pca[y==2,0],x_pca[y==2,1],s=50,c='green',label=2)
plt.scatter(model.cluster_centers_[0,0],model.cluster_centers_[0,1],s=50,c='black',label=0)
plt.legend()
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('Spotify Songs Data')
plt.show()
```

